Shijia Liu

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Professional Summary

Experienced researcher and engineer with over 7 years of expertise in natural language processing and machine learning research. Proven ability to utilize NLP tools to analyze historical texts and discover their social implications. Also with software development experience in finance. Skilled in machine learning model prototyping, data analysis, and programming.

Professional Experience

Doctoral Researcher **Boston, MA**
NULab for Digital Humanities and Computational Social Science *January 2020 - Present*
– Conducted original research on historical texts, such as 19th century American newspapers, with the goal of identifying the social implications underneath and how they correlate to historical events.
– Developed language models that help the spelling process in brain-computer interface (BCI) systems.
– Worked closely with sociologists, historians, and medical professionals.

Applied Scientist Intern at Amazon **Cambridge, MA**
Alexa AI Group *June 2022 - September 2022*
– Explored continual learning with noisy labeled data for natural language understanding tasks.
– Explored variations of EWC and self-replay methods.

Research Intern **Baltimore, MD**
JHU Center for Language and Speech Processing *May 2018 - May 2019*
– Conducted original research on the Chinese classifier system. Analyzed the correlations between classifiers and associated nouns & adjectives.
– Worked closely with linguists and computer scientists.

Financial Software Developer at Bloomberg L.P. **New York, NY**
Electronic Order Routing Group *Oct 2014 - June 2017*
– Designed and developed backend applications for the company's next-generation real-time electronic trading platform.
– Over ~100,000 trading orders from ~20,000 users flowing through the applications every day.
– Used C++, C and Python in software development. Programmed in multithreading environment.

Software Engineer II Intern at Cisco Systems, Inc **San Jose, CA**
Data Center Group *June 2013 - August 2013*
– Developed Python scripts to automatically diagnose and extract critical information from core dump files. This helped accelerate the testing process for the QA team.
– Developed Python scripts to automatically generate diagrams of UCS server clusters.

Education

Northeastern University, Ph.D. in Computer Science **Boston, MA**
Advisor: David A. Smith *December 2026 (Passed Defense)*
Specialization: Natural Language Processing, Machine Learning, Computational Social Science

Johns Hopkins University, M.S. in Computer Science — **Baltimore, MD**
Specialization: Natural Language Processing, Machine Learning — *May 2019*

Stanford University, M.S. in Electrical Engineering — **Stanford, CA**
Specialization: Optimization, Information Theory — *June 2014*

UCLA, B.S. in Electrical Engineering (Summa Cum Laude)
UCLA, B.S. in Physics (Summa Cum Laude) — **Los Angeles, CA**
Specialization: Signal Processing, Communication Systems — *June 2012*

Skills

– Programming Languages: Python, C++, C, Java, R
– Data Analysis: Pandas, Matlab, SQL, Excel
– Operating Systems: Linux, Windows
– Deep Learning Framework: PyTorch, Hugging Face, Scikit-Learn

Publications

Changes in the Coverage of East Asians in Historical American Print Media. Samir Rangwalla, Shijia Liu. In *Computational Humanities Research 2027 (CHR 2027)*

Tracing Temporal Changes of Racial Terror Accounts in Historical U.S. Newspapers (proposal). Shijia Liu, David A. Smith. In *New Directions in Analyzing Text as Data 2026 (TADA 2026)*

Through the Lens of History: Methods for Analyzing Temporal Variation in Content and Framing of State-run Chinese Newspapers. Shijia Liu, David A. Smith. In the *Proceedings of the 2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL 2025)* [ACL]

Tracing Accounts of Racial Terror in Historical Newspapers (abstract). Shijia Liu, David A. Smith. In *New Directions in Analyzing Text as Data 2023 (TADA 2023)*

Adapting Transformer Language Models for Predictive Typing in Brain-Computer Interfaces. Shijia Liu, David A. Smith. *(2023)* [arXiv]

Detecting de minimis Code-Switching in Historical German Books. Shijia Liu, David A. Smith. In the *Proceedings of the 28th International Conference on Computational Linguistics (COLING 2020)* [ACL]

Measuring the Similarity of Grammatical Gender Systems by Comparing Partitions. Arya D. McCarthy, Adina Williams, Shijia Liu, David Yarowsky, Ryan Cotterell. In the *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP 2020)* [ACL]

On the Idiosyncrasies of the Mandarin Chinese Classifier System. Shijia Liu, Hongyuan Mei, Adina Williams and Ryan Cotterell. In the *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT 2019)* [ACL]